CBSE CLASS X • INFORMATION TECHNOLOGY (SUBJECT CODE 402)

UNIT 2: ELECTRONIC SPREADSHEET (ADVANCED) — CHAPTER 6

LINKING SPREADSHEET DATA — STRUCTURED NOTEBOOK & REVISION NOTES

Syllabus Sub-Unit 3 Complete Theory • Cross-Sheet Referencing • Hyperlinks • External Web Data • Registered Data Sources • Zero Over-Shortening

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1. Need for Multiple Worksheets & Workbook Architecture

While handling real-world organizational data, a single worksheet is insufficient. Splitting records across multiple worksheets ensures structured data management and professional reporting:

- **Organizational Clarity:** Enables segregating distinct regional offices (e.g., maintaining individual expense sheets for *Branch1*, *Branch2*, *Branch3*) or keeping separate transaction ledgers for different bank accounts.
- **Consolidation & Centralized Analysis:** A master summary sheet can dynamically reference, combine, and calculate totals across all individual departmental worksheets.
- **Default Workbook Setup:** By default, a new workbook in LibreOffice Calc opens with sheets named **Sheet1** (or 3 sheets: *Sheet1*, *Sheet2*, *Sheet3* in standard configurations). These are managed via **Sheet Tabs** located at the bottom left corner.
- **Active Sheet Tab:** The sheet currently in view is shown as the active tab. Clicking any sheet tab instantly makes it active.
- **File Extension:** Workbooks created in LibreOffice Calc are saved with the standard OpenDocument Spreadsheet extension: **.ods**.

2. Setting Up & Managing Multiple Sheets (Complete Action Steps)

A. Inserting Sheets:

Step 1: Select the sheet that will be positioned next to or after the new sheet.

Step 2: Choose any of the following 3 methods:
- *Menu Bar:* Choose **Sheet** → **Insert Sheet...**.
- *Context Menu:* Right-click on any Sheet Tab at the bottom and select **Insert Sheet...**.
- *Plus Icon:* Click the **+ (Add Sheet)** icon located just before the sheet tabs.

Step 3: In the *Insert Sheet* dialog box:
- Set **Position:** Choose either *Before current sheet* or *After current sheet*.
- Set **Sheet:** Select *New sheet*, specify the *No. of sheets*, and enter a custom *Name* (e.g., *Sales_Data*).
- *From File Option:* Select *From file*, browse to select an external spreadsheet, and check **Link** if live updates are desired.

Step 4: Click **OK**.

B. Renaming a Sheet:

Methods: 1. **Double-click** directly on the sheet tab at the bottom, OR 2. Right-click the sheet tab and select **Rename Sheet...**, OR 3. Choose **Sheet** → **Rename Sheet...** from the Menu bar.

Action: In the pop-up *Rename Sheet* dialog, delete the old label, type the new name (e.g., *Q1_Summary*), and click **OK**.

C. Deleting a Sheet:

Right-click on the tab of the sheet to delete → Select **Delete Sheet...** (or select the sheet and choose **Sheet** → **Delete Sheet...** from Menu bar) → Click **Yes** in the confirmation dialog box.

3. Creating Cell References to Other Sheets (Qualified Referencing)

Definition: A **Cell Reference** refers to a cell or range of cells whose data values are used in formulas. To reference a cell residing on a *different sheet* in the same workbook, Calc requires a **qualified cell reference**.

- **The Two Essential Parts of a Qualified Reference:**
  1. **Sheet Name:** The exact title of the source sheet followed by a **dot (.) separator**.
  2. **Cell Address:** The column letter and row number identifying the cell coordinates.
  - *Standard Syntax:* SheetName.CellAddress (e.g., Sheet2.A1 or $Sheet2.$A$1).
- **Handling Spaces in Sheet Names:** If a sheet name contains blank spaces, enclose the sheet title in single quotation marks:
  - Example: 'Sales Data'.B4 or 'My Sheet'.C10.
- **A. Creating Reference via Keyboard:** Directly type the expression into the formula bar. For example, to add cell *A1* of *Sheet2* and cell *B4* of *Sheet3* inside cell *A1* of *Sheet1*, type:
  =Sheet2.A1 + Sheet3.B4
  Whenever values in *Sheet2.A1* or *Sheet3.B4* are modified, Calc automatically updates the total in *Sheet1.A1*.
- **B. Creating Reference via Mouse (Point-and-Click):**
  1. Click destination cell (e.g., *Sheet5.C4*) and type **=** (or click **=** icon on Formula Bar).
  2. Type opening expression: e.g., C2 - (.
  3. Click the **Sheet3** tab at the bottom; click cell **B15**. Calc automatically appends Sheet3.B15 to the formula.
  4. Type the mathematical operator: **+** .
  5. Click the **Sheet4** tab at the bottom; click cell **B9**. Calc appends Sheet4.B9.
  6. Type the closing bracket) and press **Enter**. Focus returns to the target cell with the evaluated sum.

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4. Hyperlinks in Spreadsheets (Concepts, Types & Dialog Options)

Definition: A **Hyperlink** in a spreadsheet is an interactive navigational link that directly opens an external file, web address, or navigates to a specific internal sheet, cell range, or bookmark within the workbook when clicked (or **Ctrl + Clicked**).

- **Prevents Document Bloat:** Keeps workbook size small by linking to external large resources instead of pasting heavy text/tables.
- **Rapid Sheet Navigation:** Allows users to jump directly between different sheets or database ranges in large multi-sheet models.
- **Insertion Shortcut & Menu:** Choose **Insert** → **Hyperlink...** (or press **Ctrl + K** / click *Hyperlink* icon on Standard Toolbar).

The Four Hyperlink Categories (Left Pane of Hyperlink Dialog):

Category	Icon	Destination & Functional Scope
1. Internet	Globe icon	Links to websites, World Wide Web URLs (HTTP/HTTPS), or FTP server addresses.
2. Mail & News	Envelope icon	Creates a clickable email link (mailto) with pre-filled recipient address and subject line.
3. Document	Document page	Links to an existing external file on the computer OR navigates to an internal target in the **current document**.
4. New Document	Blank page + star	Creates a link that generates a brand-new file immediately or sets it to be edited later.

Detailed Settings in Document Hyperlinks:

- **Path Field:** Click the folder icon to select an external file. Leave blank if linking to a location inside the *current document*.
- **Target in Document:** Click the crosshair target icon to open the navigation tree. Expand **Sheets**, **Range names**, or **Database ranges**, select the target (e.g., *House_Loan_25yr*), and click **Apply**.
- **Further Settings (Form):** Choose **Text** (displays as colored underlined text) or **Button** (displays as a clickable form button control).
- **Editing / Removing Hyperlinks:**
  - *To Edit:* Select cell without triggering the link → Choose **Insert** → **Hyperlink...** → Modify path or text → Click **Apply**.
  - *To Remove:* Right-click the hyperlinked cell → Select **Remove Hyperlink** from context menu.

5. Relative vs. Absolute Hyperlinks (High-Yield Concept & Rules)

A critical architectural distinction governs how file paths are stored and resolved across computers:

Feature / Aspect	Relative Hyperlink	Absolute Hyperlink
Starting Point	Starts from the **current location / directory** of the active open document.	Starts from the **top-most root directory** or drive letter (e.g., D:\...).
Path Notation	• Single dot (**.**) = current folder (e.g., .\data.ods). • Double dots (**..**) = parent folder (e.g., ..\ss1.ods).	Full unbroken path string: file:///D:/CBSE/IT402/Spreadsheets/Report.ods
Portability Across Systems	**Highly Portable:** Links remain functional when the entire project folder is shared, emailed, or moved to another PC.	**Breaks on Other PCs:** Fails if transferred to a computer that lacks the identical drive letter or folder hierarchy.
When Link Breaks	Breaks *only if the source and target files change position relative to each other*.	Breaks if the target file is renamed, moved, or accessed from a different drive.
Calc Display Rule	Can be saved relatively, but Calc displays URLs in **absolute form** in the dialog box.	Always displayed in full absolute format starting with file:///.

6. Linking to External Data (Web Pages & HTML Tables)

Calc allows inserting and maintaining live links to tables hosted on HTML pages or external spreadsheets. When the web page updates, Calc refreshes the data automatically.

- **Role of External Data Dialog Box:** Allows entering the source URL/file path, selecting specific named tables (e.g., HTML_1, HTML_all), and establishing an automated background refresh frequency (**Update every 'n' seconds**).
- **Step-by-Step Procedure to Link External Web Data:**
  1. Open the target Calc worksheet and select the anchor starting cell.
  2. Choose **Sheet** → **External Links...** from the Menu bar. The *External Data* dialog opens.
  3. In the **URL of External Data Source** field, enter the web address or click **Browse...** to select an HTML file. Press **Enter**.
  4. In the *Import Options* prompt, leave language locale as *Automatic* and click **OK**.
  5. In the **Available Tables/Ranges** listbox, select the desired table(s) (e.g., *HTML_1*).
  6. Check the **'Update every'** checkbox and enter a refresh interval in seconds (e.g., 60 seconds) for automatic live updates.
  7. Click **OK**. The table is inserted and dynamically synchronized with the source.

7. Linking to Registered Data Sources in LibreOffice Calc

- **Definition:** A **Registered Data Source** refers to an external database (such as a LibreOffice Base database file in **.odb** format, or external MySQL/PostgreSQL databases configured in LibreOffice) that has been added to LibreOffice's global database catalog. Once registered, all Calc spreadsheets can access and query its tables directly.
- **Viewing Shortcut (F4):** Press **F4** (or choose **View** → **Data Sources**) to toggle open the **Data Source View pane** above the spreadsheet columns.
- **Data Explorer Navigation:** The left panel displays all registered databases (including built-in *Bibliography*). Click **+** next to *Tables* and double-click a table to display its records in the right-hand data grid.
- **Inserting Records into the Sheet:**
  - *Drag and Drop:* Drag individual field headers or data records directly onto spreadsheet cells.

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• *Data to Text Button:* Select rows by clicking row header buttons in the grid → Click the **Data to Text** icon on the toolbar above the pane. Calc copies the selected records into the active worksheet with column headers.

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